BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: DIK-HSSC-XI-2021 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 18 | **Paper Code:** DIK-HSSC-XI-2021 (A)

Q1. Choose the correct option i.e. A,B,C,D and fill it up in the correct bubble in the given MCQs sheet.

Qi. The efficiency of a reaction can be determined from _____

(A) Actual yield
(B) Theoretical yield
(C) Percentage yield
(D) Both A and B

Qii. The splitting of spectral lines in magnetic field is _____

(A) Auf Bau Principal
(B) Stark's effect
(C) Zeeman effect
(D) Compton effect

Qiii. If the mass of electron is reduced to half; the Rydberg's Constant becomes_____

(A) Half
(B) Double
(C) Quadruple
(D) One fourth

Qiv. Bond angle in NH_3 is_____

(A) 90°
(B) 104.5°
(C) 107.5°
(D) 109.5°

Qv. The bond order in O_2 is_____

(A) Zero
(B) One
(C) Two
(D) Three

Qvi. The value of ideal gas constant in dm^3 torr.k^{-1} mol^{-1} is _____

(A) 0.0821
(B) 1.98722
(C) 62.364
(D) 8.3143

Qvii. Which of the following does not obey law of conservation of mass?

(A) Chemical equation
(B) Absolute zero
(C) Stiochenmetry
(D) Evaporation

Qviii. Greater number of hydrogen bonds are present in _____

(A) HF
(B) H_2O
(C) HF
(D) CH_4

Qix. A pair of compound which are isomorphic in nature is___

(A) NaCl & KNO_3
(B) MgO & ZnO
(C) $NaNo_3$ & CdS
(D) NaF & $CaCO_3$

Qx. The parameters of unit cell are_____

(A) 4
(B) 5
(C) 6
(D) 7

Qxi. $SO_2(g) + O_2(g) \rightarrow SO_3(g)$ ΔH = -ve In the above reaction production of SO_3 can be increased by:

(A) Increasing temperature
(B) Adding more SO_3
(C) Adding more SO_2
(D) Removing O_2

Qxii. Endothermic reactions are favoured in forward direction by:

(A) Cooling
(B) Heating
(C) Freezing
(D) Adding a catalyst

Qxiii. Very large Ka value means the substance is _____

(A) Strong acid
(B) Weak acid
(C) Strong base
(D) Weak base

Qxiv. In 2nd order reaction, by doubling concentration of reactants, rate of reaction_____

(A) Doubles
(B) Quadruples
(C) Halves
(D) No change

Qxv. Elevation of boiling point is a _____ property.

(A) Additive
(B) Colligative
(C) Constitutive
(D) None

Qxvi. If the mole fraction of a component in a solution is 0.4 then the mole fraction of other component is _____

(A) Zero
(B) 1
(C) 0.4
(D) 0.6

Qxvii. ΔH is always greater than zero for _____

(A) Electron affinity
(B) Combustion
(C) Ionization energy
(D) Neutralizatioin

Qxviii. A cathode has the reduction potential_____

(A) Less than anode
(B) More than anode
(C) Same as that of anode
(D) Zero

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 67

Q2. (Marks 40) Answer any TEN questions of the fol. All questions carry equal marks: [40]

i. Differentiate between limiting and excess reagents.

ii. Discuss SP^2 hybridization in the structure of ethene.

iii. Why the bond angle in NH_3 is 107.5° but in H_2O it is 104.5°?

iv. State Daltons Law of partial pressure. Write down its mathematical form.

v. How vacuum distillation is used to avoid decomposition of thermally unstable compounds?

vi. Why crystalline solids are anisotropic in nature? Justify.

vii. The equilibrium mixture contains 1 mole of PCl_5, 0.3 mole of PCl_3 and 0.08 mole of Cl_2 in 10L flask. Calculate Kc. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

viii. Prove that pH + poH = 14.

ix. In the light of collision theory of reaction rate give reason of increase in the rate of reaction by the increase of temperature.

x. In an experiment 6.4g of an organic compound was dissolved in 25.6g of benzene and 1°C depression of freezing point was observed. Calculate molecular mass of the substance (Kf of benzene = 5.12)

xi. Which one of the molarity and molality depends upon temperature? Write down the reason.

xii. Prove that $W = P\Delta V$.

xiii. Why SHE acts as anode with Cu electrode but acts as cathode when connected to Zn electrode?

SECTION – C

Maximum Marks: 27

> Note: Note: Attempt any Three Questions. All questions carry equal marks:

Q.3 a. Calculate percentage yield if 7.2g of hydrogen gas is produced actually when 5 moles of zinc is consumed in the reaction. $Zn(s) + 2HCl(or) \rightarrow ZnCl_2(or) + H_2(g)$

b. Discuss different types of overlapping of orbitals that leads to sigma (δ) and Pi (π) bond formation.

Q.4 a. State Graham's Law of diffusion and effusion. How it is experimentally verified?

b. Write down applications of hydrogen bonding.

Q.5 a. What is Le-Chatliers Principles? Discuss the effect of temperature on equilibrium of exothermic and endothermic reactions with examples.

b. What is enthalpy of a system? Prove that $\Delta H = q_p$

Q.6 a. Define Buffer solution. How acidic Buffer is made and how it resist the change in pH by addition of strong acid and strong base

b. Balance the following equation by Ion Electron method: $I^- + NO_2 \rightarrow I_2 + NO$ (Acidic Medium)